

Decent Coaching Classes

X SB

Mind Your Math - V

ANSWER KEY

Time: 1 Hour

Total Marks: 30

Note: Marks for each question are shown in []. Any other correct method should be given full credit. Answers in bold are the expected fills / final answers.

Q.1 (A) Select the correct alternative

4 Marks

- (1) **(C) 9 : 25** Ratio of areas of similar triangles = (ratio of corresponding sides)² = (3/5)² = 9/25 [1]
- (2) **(B) 7√3 cm** ∠N = 60°, so LM (opposite 60°) = (√3/2) × hypotenuse LN = (√3/2) × 14 = 7√3 [1]
- (3) **(B) 7** Scale factor 7/4 > 1, so 7 points (the larger of 7 and 4) are marked; join B₄ to C and draw a parallel through B₇ [1]
- (4) **(B) √3** Slope m = tan θ = tan 60° = √3 [1]

Q.1 (B) Solve any two

4 Marks

- (1) $d(P, Q) = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$
 $= \sqrt{[(5 + 1)^2 + (-6 - 2)^2]} = \sqrt{(36 + 64)} = \sqrt{100} = \mathbf{10}$ [2]
- (2) seg PQ || side MN ∴ LP / PM = LQ / QN (Basic proportionality theorem) [2]
∴ 6 / 4 = 9 / QN ∴ QN = (9 × 4) / 6 = **6**
- (3) Longest side = 17. Check: 8² + 15² = 64 + 225 = 289 and 17² = 289 [2]
∴ 8² + 15² = 17², so by the converse of Pythagoras theorem the triangle is **right angled**.

Q.2 (A) Activities (any two)

4 Marks

- (1) Ray BD bisects ∠ABC. [2]
∴ AB / BC = **AD** / **DC** (Angle bisector theorem)
∴ 9 / 12 = AD / DC ∴ AD / DC = 3 / **4**
Let AD = 3x, then DC = **4x**
AD + DC = AC ∴ 3x + **4x** = 14 ∴ x = **2**
∴ AD = **6** cm and DC = **8** cm.
- (2) In right angled ΔPQR, seg QS ⊥ hypotenuse PR. [2]
∴ QS² = **PS** × **SR** (Theorem of geometric mean)
∴ QS² = **4** × **9** = **36** ∴ QS = **6** cm
- (3) Let A(x₁, y₁) = A(1, 4), B(x₂, y₂) = B(7, -2) and m : n = 1 : 2. [2]
By section formula,
 $x = (m x_2 + n x_1) / (m + n) = (1 \times 7 + 2 \times \mathbf{1}) / (1 + \mathbf{2}) = \mathbf{9} / 3 = \mathbf{3}$
 $y = (m y_2 + n y_1) / (m + n) = (1 \times \mathbf{-2} + 2 \times 4) / (1 + 2) = \mathbf{6} / 3 = \mathbf{2}$
∴ Coordinates of point P are (**3**, **2**).

Q.2 (B) Solve any four

8 Marks

- (1) ΔABC ~ ΔPQR ∴ A(ΔABC) / A(ΔPQR) = AB² / PQ² (Theorem of areas of similar triangles) [2]
∴ 45 / 80 = 6² / PQ² ∴ PQ² = (36 × 80) / 45 = 64 ∴ PQ = **8 cm**
- (2) Line l || line m || line n ∴ AB / BC = PQ / QR (Property of three parallel lines and their transversals) [2]
∴ 6 / 9 = 8 / QR ∴ QR = (8 × 9) / 6 = **12**
- (3) Seg AM is a median ∴ M is the midpoint of BC ∴ BM = 6. By Apollonius theorem, [2]
AB² + AC² = 2AM² + 2BM² = 2(5)² + 2(6)² = 50 + 72 = **122**
- (4) Let ΔABC be equilateral with side 12 cm and seg AD ⊥ BC. Then D is the midpoint of BC, so BD = 6 cm. [2]
In right angled ΔADB, AD² = AB² - BD² = 144 - 36 = 108 ∴ AD = √108 = **6√3 cm**
(Or by 30°-60°-90° property: AD = (√3/2) × 12 = 6√3 cm)

- (5) Centroid $G(x, y) = ((x_1 + x_2 + x_3)/3, (y_1 + y_2 + y_3)/3)$ [2]
 $= ((-5 + 3 + 5)/3, (2 + 7 + 0)/3) = (3/3, 9/3) = \underline{(1, 3)}$

Q.3 Solve any two

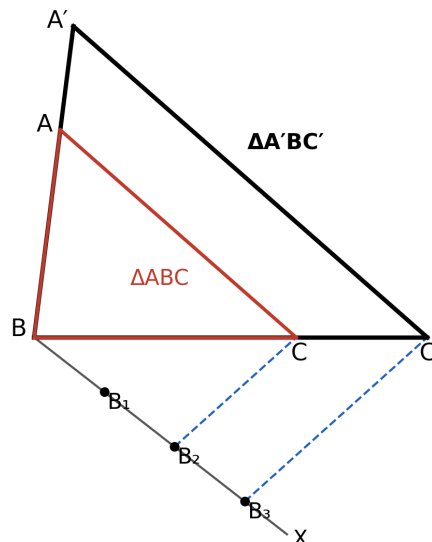
6 Marks

- (1) In $\triangle ABC$, seg $DE \parallel$ side $BC \therefore AD / DB = AE / EC$ (Basic proportionality theorem) [3]
 $\therefore x / (x - 2) = (x + 2) / (x - 1)$
 $\therefore x(x - 1) = (x + 2)(x - 2) \therefore x^2 - x = x^2 - 4 \therefore -x = -4 \therefore x = \underline{4}$
 (Check: $AD = 4, DB = 2, AE = 6, EC = 3$ and $4/2 = 6/3 = 2$)
- (2) Let $\square PQRS$ be a rhombus with diagonals $PR = 30$ cm and $QS = 16$ cm meeting at T . [3]
 Diagonals of a rhombus bisect each other at right angles $\therefore PT = 15$ cm, $QT = 8$ cm, $\angle PTQ = 90^\circ$.
 In right angled $\triangle PTQ$, $PQ^2 = PT^2 + QT^2 = 225 + 64 = 289 \therefore$ side $PQ = \underline{17 \text{ cm}}$
 Perimeter $= 4 \times \text{side} = 4 \times 17 = \underline{68 \text{ cm}}$
- (3) $P(2, 1), Q(4, k), R(6, 7)$. Slope of line $PQ = (k - 1) / (4 - 2) = (k - 1) / 2$ [3]
 Slope of line $QR = (7 - k) / (6 - 4) = (7 - k) / 2$
 P, Q, R are collinear $\therefore$ slope of $PQ =$ slope of $QR \therefore (k - 1) / 2 = (7 - k) / 2$
 $\therefore k - 1 = 7 - k \therefore 2k = 8 \therefore k = \underline{4}$

Q.4 Solve any one

4 Marks

- (a) Construction of $\triangle ABC$ and $\triangle A'BC'$ (sides $3/2$ times). [4]



Step 1: Draw $\triangle ABC$ with $BC = 5$ cm, $AB = 4$ cm (arc from B) and $AC = 6$ cm (arc from C).

Step 2: Draw ray BX making an acute angle with BC , on the side opposite to A .

Step 3: Since $3/2 > 1$, mark 3 points B_1, B_2, B_3 on ray BX such that $BB_1 = B_1B_2 = B_2B_3$.

Step 4: Join B_2 to C . Draw a line through B_3 parallel to B_2C , meeting ray BC at C' .

Step 5: Draw a line through C' parallel to CA , meeting ray BA at A' .

$\triangle A'BC'$ is the required triangle, with $BC' = 7.5$ cm, $BA' = 6$ cm and $A'C' = 9$ cm.

Marking: $\triangle ABC$ (1) + ray & equal parts (1) + parallel lines (1) + final triangle (1)

- (b) Given: $\triangle ABC \sim \triangle PQR$. To prove: $A(\triangle ABC) / A(\triangle PQR) = AB^2 / PQ^2 = BC^2 / QR^2 = AC^2 / PR^2$ [4]

Construction: Draw seg $AD \perp$ side BC and seg $PS \perp$ side QR . (Figure and given/to prove: 1)

Proof: $A(\triangle ABC) / A(\triangle PQR) = (BC \times AD) / (QR \times PS) = (BC / QR) \times (AD / PS) \dots\dots (I)$ (1)

In $\triangle ABD$ and $\triangle PQS$: $\angle B = \angle Q$ ($\triangle ABC \sim \triangle PQR$) and $\angle ADB = \angle PSQ = 90^\circ$

$\therefore \triangle ABD \sim \triangle PQS$ (AA test) $\therefore AD / PS = AB / PQ \dots\dots (II)$ (1)

$\triangle ABC \sim \triangle PQR \therefore AB / PQ = BC / QR = AC / PR \dots\dots (III)$

From (I), (II) and (III): $A(\triangle ABC) / A(\triangle PQR) = (AB / PQ) \times (AB / PQ) = AB^2 / PQ^2$

$\therefore A(\triangle ABC) / A(\triangle PQR) = AB^2 / PQ^2 = BC^2 / QR^2 = AC^2 / PR^2$ (1)